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analyst@19f03f04281f:~$
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analyst@19f03f04281f:~$ sudo ifconfig
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```
analyst@19f03f04281f:~$ sudo ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 172.18.0.2 netmask 255.255.0.0 broadcast 172.18.255.255
    ether 02:42:ac:12:00:02 txqueuelen 0 (Ethernet)
    RX packets 783 bytes 14803075 (14.1 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 442 bytes 45730 (44.6 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 73 bytes 10489 (10.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 73 bytes 10489 (10.2 KiB)
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1.eth0 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
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analyst@19f03f04281f:~$ sudo tcpdump -i eth0 -v -c5

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analyst@19f03f04281f:~$ sudo tcpdump -i eth0 -v -c5
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
07:42:58.452399 IP (tos 0x0, ttl 64, id 62078, offset 0, flags [DF], proto TCP (6), length 122)
    19f03f04281f.5000 > nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46512: Flags [P.], cksum 0x58e7 (incorrect -> 0xb314), seq 1810259298:1810259368, ack
3275623552, win 499, options [nop,nop,TS val 2402821268 ecr 308966142], length 70
07:42:58.452558 IP (tos 0x0, ttl 63, id 46585, offset 0, flags [DF], proto TCP (6), length 52)
    nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46512 > 19f03f04281f.5000: Flags [.], cksum 0x8072 (correct), ack 70, win 507, options [nop,nop,TS val 308
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34235092, win 499, options [nop,nop,TS val 2402821269 ecr 308966142], length 70
07:42:58.452706 IP (tos 0x0, ttl 63, id 2429, offset 0, flags [DF], proto TCP (6), length 52)
    nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46568 > 19f03f04281f.5000: Flags [.], cksum 0x2261 (correct), ack 70, win 507, options [nop,nop,TS val 308
966168 ecr 2402821269], length 0
07:42:58.545292 IP (tos 0x0, ttl 64, id 10839, offset 0, flags [DF], proto UDP (17), length 70)
    19f03f04281f.54919 > metadata.google.internal.domain: 60184+ PTR? 85.0.17.172.in-addr.arpa. (42)
5 packets captured
12 packets received by filter
0 packets dropped by kernel
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tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
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analyst@19f03f04281f:~$ curl opensource.google.com
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analyst@19f03f04281f:~$ curl opensource.google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="https://opensource.google/">here</A>.
</BODY></HTML>
analyst@19f03f04281f:~$ 9 packets captured
10 packets received by filter
0 packets dropped by kernel
```

```
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6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
analyst@19f03f04281f:~$ sudo tcpdump -i eth0 -v -c5
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
07:42:58.452399 IP (tos 0x0, ttl 64, id 62078, offset 0, flags [DF], proto TCP (6), length 122)
    19f03f04281f.5000 > nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46512: Flags [P.], cksum 0x58e7 (incorrect -> 0xb314), seq 1810259298:1810259368, ack
3275623552, win 499, options [nop,nop,TS val 2402821268 ecr 308966142], length 70
07:42:58.452558 IP (tos 0x0, ttl 63, id 46585, offset 0, flags [DF], proto TCP (6), length 52)
    nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46512 > 19f03f04281f.5000: Flags [.], cksum 0x8072 (correct), ack 70, win 507, options [nop,nop,TS val 308
966168 ecr 2402821268], length 0
07:42:58.452571 IP (tos 0x0, ttl 64, id 6227, offset 0, flags [DF], proto TCP (6), length 122)
    19f03f04281f.5000 > nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46568: Flags [P.], cksum 0x58e7 (incorrect -> 0x1401), seq 1496270167:1496270237, ack
84235092, win 499, options [nop,nop,TS val 2402821269 ecr 308966142], length 70
07:42:58.452706 IP (tos 0x0, ttl 63, id 2429, offset 0, flags [DF], proto TCP (6), length 52)
    nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46568 > 19f03f04281f.5000: Flags [.], cksum 0x2261 (correct), ack 70, win 507, options [nop,nop,TS val 308
966168 ecr 2402821269], length 0
07:42:58.545292 IP (tos 0x0, ttl 64, id 10839, offset 0, flags [DF], proto UDP (17), length 70)
    19f03f04281f.54919 > metadata.google.internal.domain: 60184* PTR? 85.0.17.172.in-addr.arpa. (42)
5 packets captured
12 packets received by filter
0 packets dropped by kernel
analyst@19f03f04281f:~$ sudo tcpdump -i eth0 -nn -c9 port 80 -w capture.pcap &
[1] 712
analyst@19f03f04281f:~$ tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
analyst@19f03f04281f:~$ curl opensource.google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
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<A HREF="https://opensource.google/">here</A>.
</BODY></HTML>
analyst@19f03f04281f:~$ 9 packets captured
10 packets received by filter
0 packets dropped by kernel
ls -l capture.pcap
```

```
/usr/bin/dbus-system (D-Bus system bus) [none]
/usr/bin/dbus-session (D-Bus session bus) [none]
analyst@19f03f04281f:~$ sudo tcpdump -i eth0 -v -c5
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
07:42:58.452399 IP (tos 0x0, ttl 64, id 62078, offset 0, flags [DF], proto TCP (6), length 122)
    19f03f04281f.5000 > nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46512: Flags [P.], cksum 0x58e7 (incorrect -> 0xb314), seq 1810259298:1810259368, ack
    3275623552, win 499, options [nop,nop,TS val 2402821268 ecr 308966142], length 70
07:42:58.452558 IP (tos 0x0, ttl 63, id 46585, offset 0, flags [DF], proto TCP (6), length 52)
    nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46512 > 19f03f04281f.5000: Flags [.], cksum 0x8072 (correct), ack 70, win 507, options [nop,nop,TS val 308
    966168 ecr 2402821268], length 0
07:42:58.452571 IP (tos 0x0, ttl 64, id 6227, offset 0, flags [DF], proto TCP (6), length 122)
    19f03f04281f.5000 > nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46568: Flags [P.], cksum 0x58e7 (incorrect -> 0x1401), seq 1496270167:1496270237, ack
    84235092, win 499, options [nop,nop,TS val 2402821269 ecr 308966142], length 70
07:42:58.452706 IP (tos 0x0, ttl 63, id 2429, offset 0, flags [DF], proto TCP (6), length 52)
    nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46568 > 19f03f04281f.5000: Flags [.], cksum 0x2261 (correct), ack 70, win 507, options [nop,nop,TS val 308
    966168 ecr 2402821269], length 0
07:42:58.545292 IP (tos 0x0, ttl 64, id 10839, offset 0, flags [DF], proto UDP (17), length 70)
    19f03f04281f.54919 > metadata.google.internal.domain: 60184+ PTR? 85.0.17.172.in-addr.arpa. (42)
5 packets captured
12 packets received by filter
0 packets dropped by kernel
analyst@19f03f04281f:~$ sudo tcpdump -i eth0 -nn -c9 port 80 -w capture.pcap &
[1] 712
analyst@19f03f04281f:~$ tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes

analyst@19f03f04281f:~$ curl opensource.google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="https://opensource.google/">here</A>.
</BODY></HTML>
analyst@19f03f04281f:~$ 9 packets captured
10 packets received by filter
0 packets dropped by kernel
ls -l capture.pcap
-rw-r--r-- 1 tcpdump tcpdump 1401 Sep 23 07:53 capture.pcap
[1]+  Done                  sudo tcpdump -i eth0 -nn -c9 port 80 -w capture.pcap
analyst@19f03f04281f:~$
```

```
/usr/bin/dbus-system (D-Bus system bus) [none]
/usr/bin/dbus-session (D-Bus session bus) [none]
analyst@19f03f04281f:~$ sudo tcpdump -i eth0 -v -c5
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
07:42:58.452399 IP (tos 0x0, ttl 64, id 62078, offset 0, flags [DF], proto TCP (6), length 122)
    19f03f04281f.5000 > nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46512: Flags [P.], cksum 0x58e7 (incorrect -> 0xb314), seq 1810259298:1810259368, ack
    3275623552, win 499, options [nop,nop,TS val 2402821268 ecr 308966142], length 70
07:42:58.452558 IP (tos 0x0, ttl 63, id 46585, offset 0, flags [DF], proto TCP (6), length 52)
    nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46512 > 19f03f04281f.5000: Flags [.], cksum 0x8072 (correct), ack 70, win 507, options [nop,nop,TS val 308
    966168 ecr 2402821268], length 0
07:42:58.452571 IP (tos 0x0, ttl 64, id 6227, offset 0, flags [DF], proto TCP (6), length 122)
    19f03f04281f.5000 > nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46568: Flags [P.], cksum 0x58e7 (incorrect -> 0x1401), seq 1496270167:1496270237, ack
    84235092, win 499, options [nop,nop,TS val 2402821269 ecr 308966142], length 70
07:42:58.452706 IP (tos 0x0, ttl 63, id 2429, offset 0, flags [DF], proto TCP (6), length 52)
    nginx-us-central1-b.c.qwiklabs-terminal-vms-prod-00.internal.46568 > 19f03f04281f.5000: Flags [.], cksum 0x2261 (correct), ack 70, win 507, options [nop,nop,TS val 308
    966168 ecr 2402821269], length 0
07:42:58.545292 IP (tos 0x0, ttl 64, id 10839, offset 0, flags [DF], proto UDP (17), length 70)
    19f03f04281f.54919 > metadata.google.internal.domain: 60184+ PTR? 85.0.17.172.in-addr.arpa. (42)
5 packets captured
12 packets received by filter
0 packets dropped by kernel
analyst@19f03f04281f:~$ sudo tcpdump -i eth0 -nn -c9 port 80 -w capture.pcap &
[1] 712
analyst@19f03f04281f:~$ tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes

analyst@19f03f04281f:~$ curl opensource.google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
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<H1>301 Moved</H1>
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</BODY></HTML>
analyst@19f03f04281f:~$ 9 packets captured
10 packets received by filter
0 packets dropped by kernel
ls -l capture.pcap
-rw-r--r-- 1 tcpdump tcpdump 1401 Sep 23 07:53 capture.pcap
[1]+  Done                  sudo tcpdump -i eth0 -nn -c9 port 80 -w capture.pcap
analyst@19f03f04281f:~$ sudo tcpdump -nn -r capture.pcap -v
```

```
172.18.0.2.49304 > 173.194.195.138.80: Flags [.], cksum 0x1d88 (incorrect -> 0x0d2b), ack 1, win 511, options [nop,nop,TS val 4081307919 ecr 225459804], length 0
07:53:09.363919 IP (tos 0x0, ttl 64, id 44476, offset 0, flags [DF], proto TCP (6), length 137)
    172.18.0.2.49304 > 173.194.195.138.80: Flags [P.], cksum 0x1ddd (incorrect -> 0x79d3), seq 1:86, ack 1, win 511, options [nop,nop,TS val 4081307919 ecr 225459804], len
    gth 85: HTTP, length: 85
    GET / HTTP/1.1
    Host: opensource.google.com
    User-Agent: curl/7.88.1
    Accept: */*
07:53:09.364251 IP (tos 0x0, ttl 126, id 0, offset 0, flags [DF], proto TCP (6), length 52)
    173.194.195.138.80 > 172.18.0.2.49304: Flags [.], cksum 0x0aba (correct), ack 86, win 1051, options [nop,nop,TS val 225459804 ecr 4081307919], length 0
07:53:09.366963 IP (tos 0x0, ttl 126, id 0, offset 0, flags [DF], proto TCP (6), length 590)
    173.194.195.138.80 > 172.18.0.2.49304: Flags [P.], cksum 0xf894 (correct), seq 1:539, ack 86, win 1051, options [nop,nop,TS val 225459807 ecr 4081307919], length 538:
    HTTP, length: 538
    HTTP/1.1 301 Moved Permanently
    Location: https://opensource.google/
    Content-Type: text/html; charset=UTF-8
    X-Content-Type-Options: nosniff
    Date: Wed, 23 Sep 2026 07:53:09 GMT
    Expires: Wed, 23 Sep 2026 08:23:09 GMT
    Cache-Control: public, max-age=1800
    Server: sffe
    Content-Length: 223
    X-XSS-Protection: 0

    <HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
    <TITLE>301 Moved</TITLE></HEAD><BODY>
    <H1>301 Moved</H1>
    The document has moved
    <A HREF="https://opensource.google/">here</A>.
    </BODY></HTML>
07:53:09.366979 IP (tos 0x0, ttl 64, id 44477, offset 0, flags [DF], proto TCP (6), length 52)
    172.18.0.2.49304 > 173.194.195.138.80: Flags [.], cksum 0x1d88 (incorrect -> 0x0aba), ack 539, win 507, options [nop,nop,TS val 4081307922 ecr 225459807], length 0
07:53:09.367226 IP (tos 0x0, ttl 64, id 44478, offset 0, flags [DF], proto TCP (6), length 52)
    172.18.0.2.49304 > 173.194.195.138.80: Flags [F.], cksum 0x1d88 (incorrect -> 0x0ab39), seq 86, ack 539, win 507, options [nop,nop,TS val 4081307922 ecr 225459807], len
    gth 0
07:53:09.367458 IP (tos 0x0, ttl 126, id 0, offset 0, flags [DF], proto TCP (6), length 52)
    173.194.195.138.80 > 172.18.0.2.49304: Flags [F.], cksum 0x0897 (correct), seq 539, ack 87, win 1051, options [nop,nop,TS val 225459808 ecr 4081307922], length 0
analyst@19f03f04281f:~$
```

```
172.18.0.2.49304 > 173.194.195.138.80: Flags [.], cksum 0xid88 (incorrect -> 0x0d2b), ack 1, win 511, options [nop,nop,TS val 4081307919 ecr 225459804], length 0
07:53:09.363919 IP (tos 0x0, ttl 64, id 44476, offset 0, flags [DF], proto TCP (6), length 137)
172.18.0.2.49304 > 173.194.195.138.80: Flags [P.], cksum 0xlddd (incorrect -> 0x79d9), seq 1:86, ack 1, win 511, options [nop,nop,TS val 4081307919 ecr 225459804], len
gth 85: HTTP, length: 85
GET / HTTP/1.1
Host: opensource.google.com
User-Agent: curl/7.88.1
Accept: */*

07:53:09.364251 IP (tos 0x0, ttl 126, id 0, offset 0, flags [DF], proto TCP (6), length 52)
173.194.195.138.80 > 172.18.0.2.49304: Flags [.], cksum 0x0aba (correct), ack 86, win 1051, options [nop,nop,TS val 225459804 ecr 4081307919], length 0
07:53:09.366963 IP (tos 0x0, ttl 126, id 0, offset 0, flags [DF], proto TCP (6), length 590)
173.194.195.138.80 > 172.18.0.2.49304: Flags [P.], cksum 0xf9b94 (correct), seq 1:539, ack 86, win 1051, options [nop,nop,TS val 225459807 ecr 4081307919], length 538:
HTTP, length: 538
HTTP/1.1 301 Moved Permanently
Location: https://opensource.google/
Content-Type: text/html; charset=UTF-8
X-Content-Type-Options: nosniff
Date: Wed, 23 Sep 2026 07:53:09 GMT
Expires: Wed, 23 Sep 2026 08:23:09 GMT
Cache-Control: public, max-age=1800
Server: sffe
Content-Length: 223
X-XSS-Protection: 0

<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
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07:53:09.366979 IP (tos 0x0, ttl 64, id 44477, offset 0, flags [DF], proto TCP (6), length 52)
172.18.0.2.49304 > 173.194.195.138.80: Flags [.], cksum 0xid88 (incorrect -> 0x0aba), ack 539, win 507, options [nop,nop,TS val 4081307922 ecr 225459807], length 0
07:53:09.367226 IP (tos 0x0, ttl 64, id 44478, offset 0, flags [DF], proto TCP (6), length 52)
172.18.0.2.49304 > 173.194.195.138.80: Flags [F.], cksum 0xld88 (incorrect -> 0x0ab9), seq 86, ack 539, win 507, options [nop,nop,TS val 4081307922 ecr 225459807], len
gth 0
07:53:09.367458 IP (tos 0x0, ttl 126, id 0, offset 0, flags [DF], proto TCP (6), length 52)
173.194.195.138.80 > 172.18.0.2.49304: Flags [F.], cksum 0x0897 (correct), seq 539, ack 87, win 1051, options [nop,nop,TS val 225459808 ecr 4081307922], length 0
analyst@19f03f04281f:~$ sudo tcpdump -nn -r capture.pcap -x
```

```
0x00e0: 3039 2047 4d54 0d0a 4578 7069 7265 733a 09.GMT..Expires:
0x00f0: 2057 6564 2c20 3233 2053 6570 2032 3032 .Wed,.23.Sep.202
0x0100: 3620 3038 3a32 333a 3039 2047 4d54 0d0a 6.08:23:09.GMT..
0x0110: 4361 6368 652d 436f 6e74 726f 6c3a 2070 Cache-Control:p
0x0120: 7562 6e69 632c 2064 6178 2d61 6765 3d31 ublic,max-age=1
0x0130: 3830 300d 0a53 6572 7665 723a 2073 6666 800..Server:sff
0x0140: 650d 0a43 6f6e 7465 6e74 2d4c 656e 6774 e..Content-Lengt
0x0150: 683a 2032 3233 0d0a 582d 5853 532d 5072 h:223..X-XSS-Pr
0x0160: 6f74 6563 7469 6f6e 3a20 300d 0a0d 0a3c otection:0....<
0x0170: 4854 4d4c 3e3c 4845 4144 3e3c 6d65 7461 HTML><HEAD><meta
0x0180: 2068 7474 702d 6571 7569 763d 2263 6f6e .http-equiv="con
0x0190: 7465 6e74 2d74 7970 6522 2063 6f6e 7465 tent-type".conte
0x01a0: 6e74 3d22 7465 7874 2f68 746d 6c3b 6368 nt="text/html;ch
0x01b0: 6172 7365 743d 7574 662d 3822 3e0a 3c54 arset=utf-8"><T
0x01c0: 4954 4e45 3e33 3031 204d 6f76 6564 3c2f TITLE>301 Moved</
0x01d0: 5449 544c 453e 3c2f 4845 4144 3e3c 424f TITLE></HEAD><B
0x01e0: 4459 3e0a 3c48 313e 3330 3120 4d6f 7665 DY>.<H1>301.Move
0x01f0: 643c 2f48 313e 0a54 6865 2064 6f63 756d d</H1>.The.docum
0x0200: 656e 7420 6861 7320 6d6f 7665 640a 3c41 ent.has.moved.<A
0x0210: 2048 5245 463d 2268 7474 7073 3a2f 2f6f .HREF="https://o
0x0220: 7065 6e73 6f75 7263 652e 676f 6f67 6c65 pensource.google
0x0230: 2f22 3e68 6572 653c 2f41 3e2e 0d0a 3c2f /*>here</A>...</
0x0240: 424f 4459 3e3c 2f48 544d 4c3e 0d0a BODY></HTML>..
07:53:09.366979 IP 172.18.0.2.49304 > 173.194.195.138.80: Flags [.], ack 539, win 507, options [nop,nop,TS val 4081307922 ecr 225459807], length 0
0x0000: 4500 0034 adbd 4000 4006 6fa5 ac12 0002 E..4..8..8.o.....
0x0010: adc2 c38a c098 0050 a29f c1ef b00c 66fc .....P.....f.
0x0020: 8010 01fb 1d88 0000 0101 080a f343 d112 .....C..
0x0030: 0d70 3e5f .p>
07:53:09.367226 IP 172.18.0.2.49304 > 173.194.195.138.80: Flags [F.], seq 86, ack 539, win 507, options [nop,nop,TS val 4081307922 ecr 225459807], length 0
0x0000: 4500 0034 adbe 4000 4006 6fa4 ac12 0002 E..4..8..8.o.....
0x0010: adc2 c38a c098 0050 a29f c1ef b00c 66fc .....P.....f.
0x0020: 8011 01fb 1d88 0000 0101 080a f343 d112 .....C..
0x0030: 0d70 3e5f .p>
07:53:09.367458 IP 173.194.195.138.80 > 172.18.0.2.49304: Flags [F.], seq 539, ack 87, win 1051, options [nop,nop,TS val 225459808 ecr 4081307922], length 0
0x0000: 4500 0034 0000 4000 7e06 df62 adc2 c38a E..4..8..b....
0x0010: ac12 0002 0050 c098 b00c 66fc a29f c1f0 .....P.....f.....
0x0020: 8011 041b 0897 0000 0101 080a 0d70 3e60 .....p>
0x0030: f343 d112 .C..
analyst@19f03f04281f:~$
```